1. Sudden Rise in Chocolate Sales During Halloween

i. What happened? (Descriptive Analytics):

Chocolate sales increased significantly during Halloween compared to previous months.

ii. Why did it happen? (Diagnostic Analytics):

Increased demand for Halloween treats, trick-or-treating, promotional offers, and festive shopping contributed to higher sales.

iii. What might happen next? (Predictive Analytics):

Chocolate sales may rise again during upcoming festivals and holidays such as Christmas and Valentine's Day.

iv. What should we do? (Prescriptive Analytics):

Increase inventory before Halloween, launch festive promotions, and optimize pricing to maximize profits.

2. Increase in Crime Rate During Lockdown

i. What happened? (Descriptive Analytics):

Recorded crime rates increased during the lockdown period.

ii. Why did it happen? (Diagnostic Analytics):

Analyze crime categories, locations, reporting patterns, economic stress, and changes in policing to identify possible causes.

iii. What might happen next? (Predictive Analytics):

Certain crime categories may increase or decrease as restrictions change, depending on economic conditions and policing patterns.

iv. What should we do? (Prescriptive Analytics):

Allocate police resources to identified high-risk areas, strengthen community support, and develop targeted crime-prevention programs.

3. Rise in New Startups After Cloud Computing Was Introduced

i. What happened? (Descriptive Analytics):

The number of new startups increased as cloud computing adoption grew.

ii. Why did it happen? (Diagnostic Analytics):

Cloud services reduced the need for expensive hardware, lowered initial infrastructure costs, and made it easier to scale applications.

iii. What might happen next? (Predictive Analytics):

Startup creation may continue to grow in areas such as Artificial Intelligence (AI), Software as a Service (SaaS), and cloud-based services.

iv. What should we do? (Prescriptive Analytics):

Entrepreneurs should use scalable cloud infrastructure, while investors and policymakers should support promising startups and digital innovation.

4. An Animation Series Became Famous Recently

i. What happened? (Descriptive Analytics):

The animation series experienced a sudden increase in viewers, social media mentions, and audience engagement.

ii. Why did it happen? (Diagnostic Analytics):

Analyze audience reviews, social media trends, influencer promotions, story quality, release timing, and viewer demographics to identify the reasons for its popularity.

iii. What might happen next? (Predictive Analytics):

The series may attract more viewers, gain new fans, and generate demand for future seasons or merchandise.

iv. What should we do? (Prescriptive Analytics):

Promote the series to similar audiences, release engaging content, improve audience interaction, and plan future episodes based on viewer preferences.

5. Sudden Increase in Customer Complaints at an E-commerce Company

i. What happened? (Descriptive Analytics):

Customer complaints increased significantly over the past month.

ii. Why did it happen? (Diagnostic Analytics):

Analyze customer feedback, delivery delays, damaged products, and payment failures to identify the main causes of the increase in complaints.

iii. What might happen next? (Predictive Analytics):

If the issues continue, customer satisfaction may decline, and more customers may stop purchasing from the platform.

iv. What should we do? (Prescriptive Analytics):

Improve delivery operations, resolve payment issues, strengthen quality checks, and provide faster customer support.